

Course Map

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1. Overview

Key point: The Course map ties every Note together. It shows where each idea sits in an ML project, how ideas connect, what to read first, and which algorithm to choose.

The Notes each teach one lesson. The Course map shows how those lessons fit together, in four views:

View	Question it answers
Pipeline map	Where does this idea sit in a real ML project?
Concept map	How is this idea connected to the others?
Learning path	Which Notes should I read first?
Algorithm chooser	Which algorithm suits my problem?

Each idea on the map is a **Concept**. Concepts already taught in a written Note are **confirmed**; the others are **draft**, shown faint, and are checked against their Video when their Note is written. So far, 78 of 136 Concepts are confirmed.

Every Note starts with a *Where this fits* box: a small Pipeline map with that Note's steps highlighted, plus what it builds on and what it leads to.

Extra: This map is generated from one data file, `course_map/concepts.yaml`. An interactive version, where you can zoom, filter by step and click a Concept to see its links and Notes, runs with `python course_map/app.py`.

2. The Pipeline map

Key point: Every ML project moves through the same steps, from framing the problem to monitoring the deployed model. Every Concept belongs to one step.

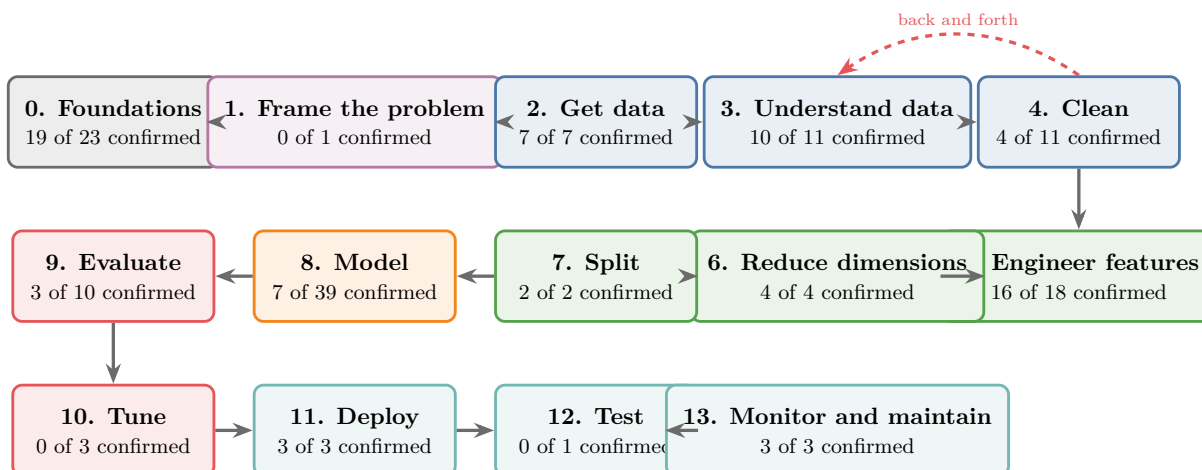


Figure 1: The Pipeline map: 14 steps of an ML project

Figure 1 shows the 14 steps. They follow the ML development life cycle: frame the problem, get and understand the data, clean it, engineer features, train and evaluate models, tune them, deploy, test and keep the model healthy.

Steps 3 (Understand data) and 4 (Clean) loop: exploring reveals what needs cleaning, and cleaning changes the data, so we explore again.

2.1 Step 0: Foundations

Concept	Taught in	Status
Machine learning	Video 1, coming, Note 2	confirmed
Data mining	Video 1, coming	draft
Artificial intelligence	Note 2	confirmed
Symbolic AI and expert systems	Note 2	confirmed
Deep learning	Note 2	confirmed
Neural networks	Note 2	confirmed
Features	Note 2 , Note 11	confirmed
Supervised learning	Note 3	confirmed
Regression problems	Note 3	confirmed
Classification problems	Note 3	confirmed
Unsupervised learning	Note 3	confirmed
Semi-supervised learning	Note 3	confirmed
Reinforcement learning	Note 3	confirmed
Batch (offline) learning	Note 4	confirmed
Online learning	Note 5	confirmed
Out-of-core learning	Note 5	confirmed
Instance-based learning	Note 6	confirmed
Model-based learning	Note 6	confirmed
Applications of ML	Video 8, coming	draft
ML development life cycle	Video 9, coming, Note 13	draft
Tensors	Note 11	confirmed
Anaconda, Jupyter and Colab	Video 12, coming	draft
Eigenvectors and eigenvalues	Note 48	confirmed

2.2 Step 1: Frame the problem

Concept	Taught in	Status
Framing an ML problem	Video 9, coming, Video 14, coming	draft

2.3 Step 2: Get data

Concept	Taught in	Status
Labelled data	Note 3 , Note 7	confirmed
Enough data	Note 7	confirmed
Sampling noise and bias	Note 7	confirmed
APIs	Note 7 , Note 17	confirmed
Web scraping	Note 7 , Note 18	confirmed
CSV files	Note 13 , Note 15	confirmed
JSON and SQL data	Note 16	confirmed

2.4 Step 3: Understand data

Concept	Taught in	Status
Imbalanced data	Video 9, coming, Video 133, coming	draft
Exploratory data analysis	Note 13 , Note 19	confirmed
Variance	Note 19 , Note 47	confirmed
Correlation	Note 19 , Note 21	confirmed
Descriptive statistics	Note 19	confirmed
Univariate analysis	Note 20	confirmed
Skewness	Note 20	confirmed
Bivariate and multivariate analysis	Note 21	confirmed
Pandas Profiling	Note 22	confirmed
Q-Q plot	Note 30	confirmed
Covariance and covariance matrix	Note 48	confirmed

2.5 Step 4: Clean

Concept	Taught in	Status
Poor-quality data	Note 7	confirmed
Missing values	Note 7 , Note 23 , Note 35 , Note 36 , Note 37	confirmed
Outliers	Note 7 , Note 20 , Note 23 , Video 41, coming	draft
Simple imputation (mean, median, mode, constant)	Note 23 , Note 36 , Note 37	confirmed
Complete case analysis	Note 35	confirmed
Missing indicator	Video 38, coming	draft
KNN imputer	Video 39, coming	draft
Iterative imputation (MICE)	Video 40, coming	draft

Concept	Taught in	Status
Z-score outlier method	Video 42, coming	draft
IQR outlier method	Video 43, coming	draft
Percentile outlier method	Video 44, coming	draft

2.6 Step 5: Engineer features

Concept	Taught in	Status
Feature scaling	Note 6 , Note 13 , Note 23 , Note 24	confirmed
Feature engineering	Note 7 , Note 23	confirmed
Feature construction and splitting	Note 7 , Note 23 , Video 45, coming	draft
Feature selection	Video 9, coming, Note 13 , Note 23 , Note 46	confirmed
One-hot encoding	Note 11 , Note 23 , Note 27	confirmed
Standardization	Note 13 , Note 24	confirmed
ML pipelines	Note 13 , Note 29	confirmed
Encoding categorical data	Note 23 , Note 26	confirmed
Binning and binarization	Note 23 , Note 32	confirmed
Feature transformation	Note 23	confirmed
Normalization	Note 25	confirmed
Ordinal and label encoding	Note 26	confirmed
Column transformer	Note 28	confirmed
Function transformer	Note 30	confirmed
Power transformer	Note 31	confirmed
Mixed variables	Note 33	confirmed
Date and time features	Note 34	confirmed
Feature importance	Video 114, coming	draft

2.7 Step 6: Reduce dimensions

Concept	Taught in	Status
Dimensionality reduction	Note 3 , Note 46	confirmed
PCA	Note 3 , Note 47 , Note 48 , Note 49	confirmed
Feature extraction	Note 23 , Note 46 , Note 47	confirmed
Curse of dimensionality	Note 46	confirmed

2.8 Step 7: Split

Concept	Taught in	Status
Train-test split	Note 13	confirmed
Data leakage	Note 13	confirmed

2.9 Step 8: Model

Concept	Taught in	Status
Clustering	Note 3 , Video 128, coming	confirmed
Anomaly detection	Note 3	confirmed
Association rule learning	Note 3	confirmed
Stochastic gradient descent	Note 5 , Video 59, coming	draft
K-nearest neighbours	Note 6 , Video 91, coming	confirmed
Logistic regression	Note 13 , Video 70, coming, Video 71, coming, Video 72, coming, Video 73, coming, Video 75, coming	draft
Multicollinearity	Note 27	confirmed
Simple linear regression	Note 50 , Video 51, coming	confirmed
Best-fit line and squared error	Note 50 , Video 51, coming	confirmed
Multiple linear regression	Video 53, coming, Video 54, coming, Video 55, coming	draft
Assumptions of linear regression	Video 56, coming	draft
Gradient descent	Video 57, coming	draft
Batch gradient descent	Video 58, coming	draft
Mini-batch gradient descent	Video 60, coming	draft
Polynomial regression	Video 61, coming	draft
Polynomial features	Video 61, coming, Video 80, coming	draft
Regularisation	Video 63, coming	draft
Ridge regression	Video 63, coming, Video 64, coming, Video 65, coming, Video 66, coming	draft
Lasso regression	Video 67, coming, Video 68, coming	draft
ElasticNet	Video 69, coming	draft
Sigmoid function	Video 74, coming	draft
Softmax regression	Video 79, coming	draft

Concept	Taught in	Status
Naive Bayes	Video 82, coming, Video 83, coming, Video 84, coming, Video 85, coming, Video 86, coming, Video 87, coming, Video 88, coming, Video 89, coming, Video 90, coming	draft
Support vector machines	Video 92, coming, Video 93, coming, Video 94, coming	draft
Kernel trick	Video 95, coming, Video 96, coming	draft
Decision trees	Video 97, coming, Video 98, coming, Video 100, coming	draft
Regression trees	Video 99, coming	draft
Ensemble learning	Video 101, coming	draft
Voting ensembles	Video 102, coming, Video 103, coming, Video 104, coming	draft
Bagging	Video 105, coming, Video 106, coming, Video 107, coming	draft
Random forest	Video 108, coming, Video 109, coming, Video 110, coming, Video 111, coming	draft
AdaBoost	Video 115, coming, Video 116, coming, Video 117, coming, Video 118, coming	draft
Boosting	Video 119, coming	draft
Gradient boosting	Video 120, coming, Video 121, coming, Video 122, coming	draft
XGBoost	Video 123, coming, Video 124, coming, Video 125, coming, Video 126, coming	draft
Stacking and blending	Video 127, coming	draft
K-means	Video 128, coming, Video 129, coming, Video 130, coming	draft
Hierarchical clustering	Video 131, coming	draft
DBSCAN	Video 132, coming	draft

2.10 Step 9: Evaluate

Concept	Taught in	Status
Overfitting	Note 7	confirmed
Underfitting	Note 7	confirmed
Accuracy	Note 13 , Video 76, coming	confirmed
Cross-validation	Note 29 , Note 30 , Video 112, coming	draft
Regression metrics	Video 52, coming	draft
Bias-variance trade-off	Video 62, coming, Video 109, coming	draft
Confusion matrix	Video 76, coming	draft
Precision, recall and F1	Video 77, coming	draft
ROC curve and AUC	Video 78, coming	draft
OOB score	Video 113, coming	draft

2.11 Step 10: Tune

Concept	Taught in	Status
Hyperparameter tuning	Video 9, coming, Video 81, coming, Video 98, coming, Video 111, coming, Video 118, coming	draft
Grid and random search	Note 29 , Video 112, coming	draft
Optuna	Video 134, coming	draft

2.12 Step 11: Deploy

Concept	Taught in	Status
Deployment	Note 4 , Note 7 , Note 13 , Note 29	confirmed
Software integration	Note 7	confirmed
Saving models with pickle	Note 13 , Note 29	confirmed

2.13 Step 12: Test

Concept	Taught in	Status
Beta and A/B testing	Video 9, coming	draft

2.14 Step 13: Monitor and maintain

Concept	Taught in	Status
Model drift	Note 4	confirmed
Retraining	Note 4 , Note 5	confirmed
MLOps and cost	Note 7	confirmed

3. The Concept map

Key point: Concepts are joined by five kinds of Link. Following the Links shows why each idea exists and what it is for.

Every Link has one of five types:

Link	Meaning	Example
needs	must be understood first	logistic regression <i>needs</i> gradient descent
is a kind of	a special case	Ridge <i>is a kind of</i> regularisation
fixes	solves a problem	regularisation <i>fixes</i> overfitting
compared with	often confused or contrasted	bagging <i>compared with</i> boosting
used in	a tool used inside something else	feature scaling <i>used in</i> KNN

The full map has 136 Concepts, too many for one page, so it is shown in six areas (Figures 2 to 7). In each figure, the area's own Concepts are large and coloured by pipeline step; Concepts from other areas that link in are small and grey. Faint dots and dotted lines are drafts.

Concept map: Foundations and framing

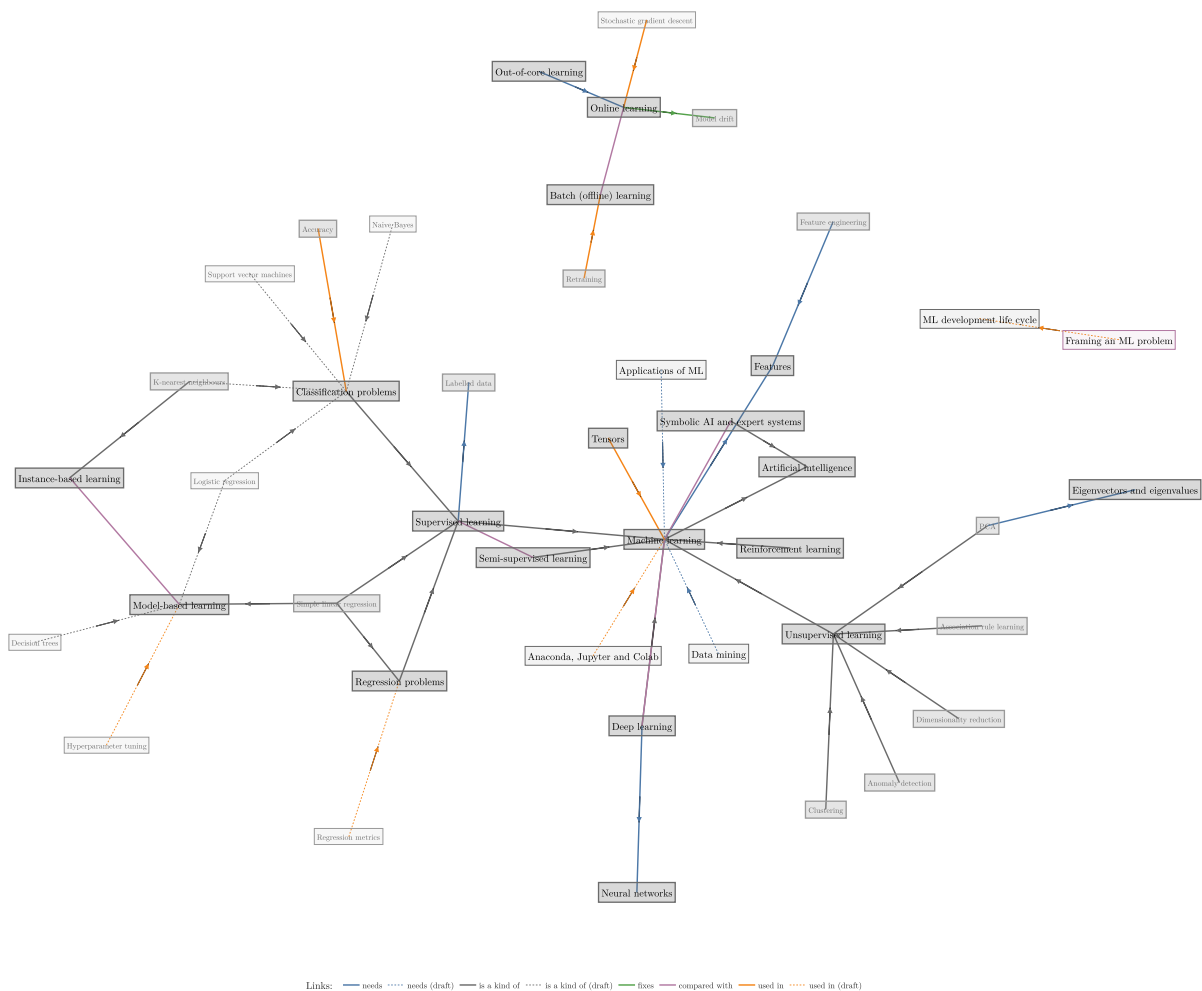
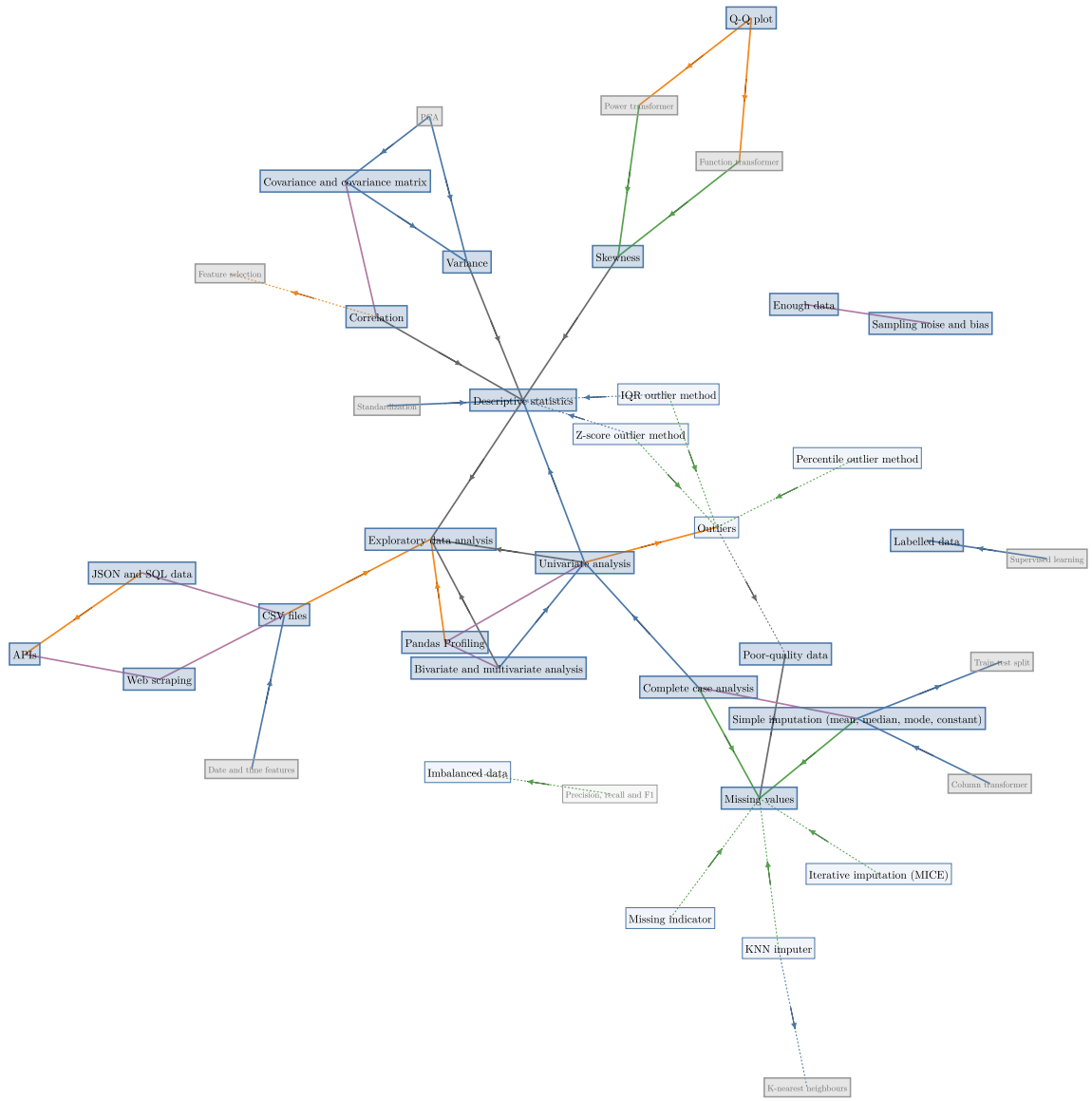


Figure 2: Concept map: Foundations and framing

Concept map: Getting, understanding and cleaning data



Links: — needs needs (draft) — is a kind of is a kind of (draft) — fixes fixes (draft) — compared with — used in used in (draft)

Figure 3: Concept map: Getting, understanding and cleaning data

Figure 4: Concept map: Features, dimensions and splitting

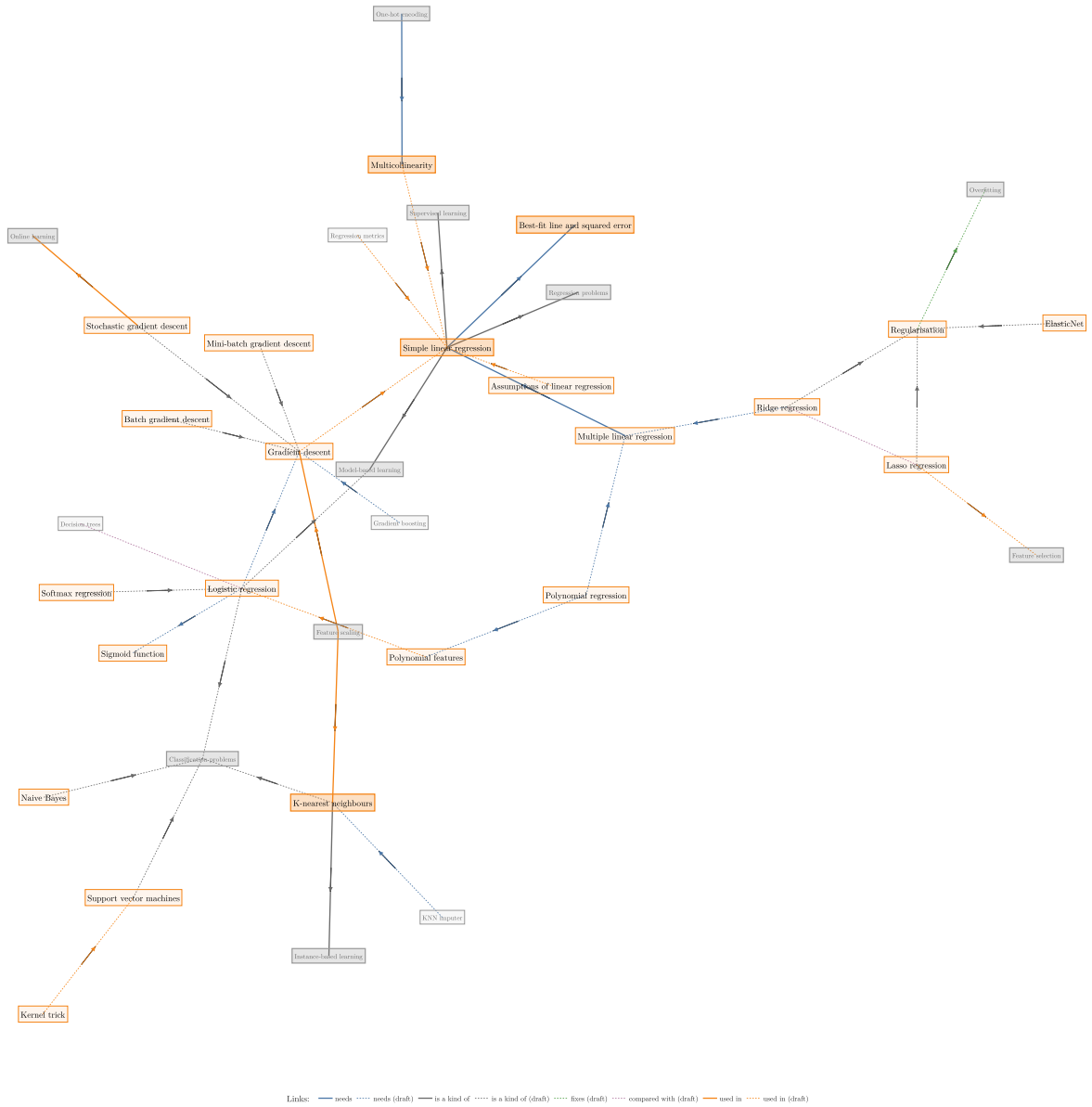


Figure 5: Concept map: Models: regression, classification and gradient descent

Concept map: Models: trees, ensembles and clustering

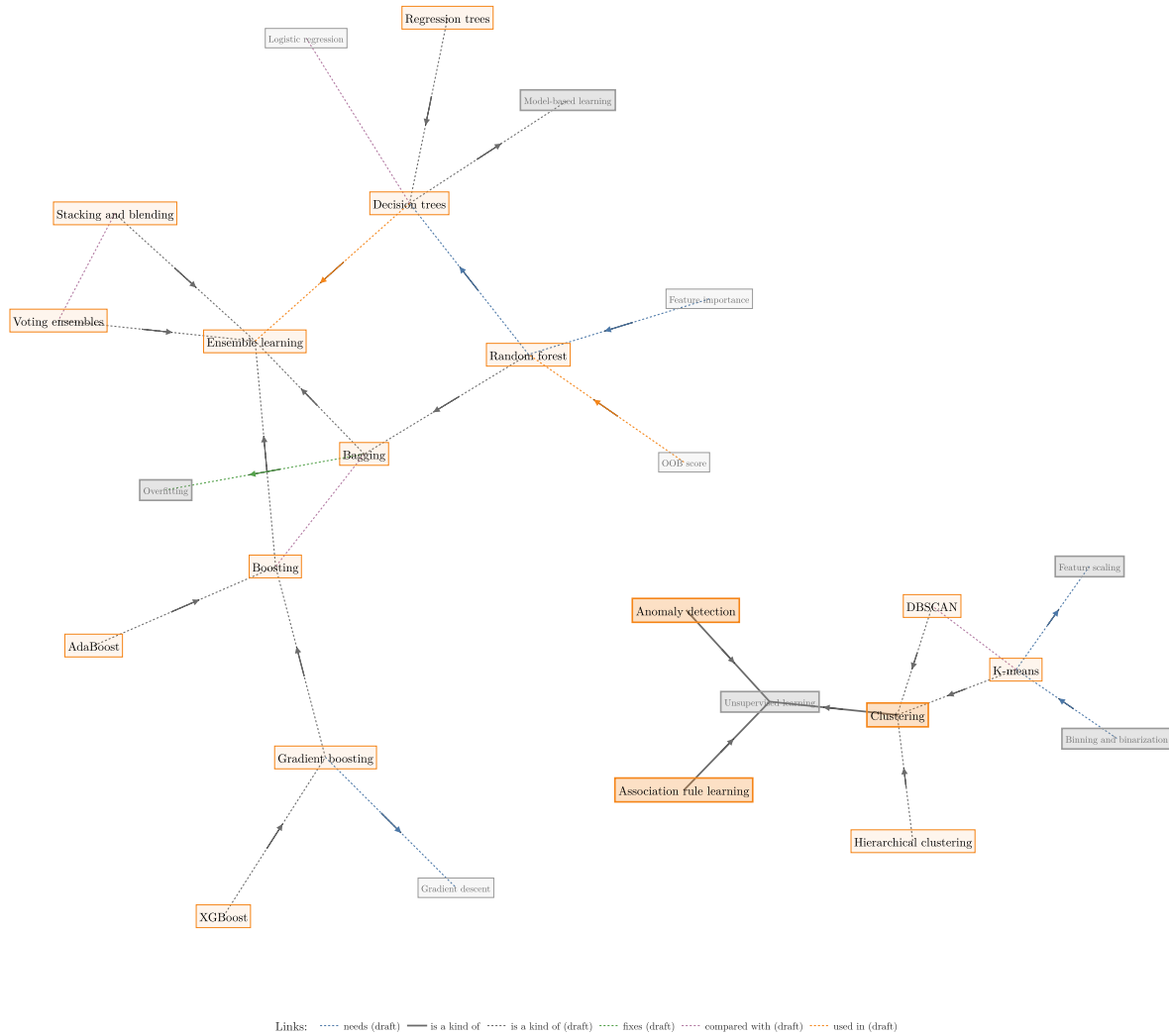


Figure 6: Concept map: Models: trees, ensembles and clustering

```

graph TD
    A[Imbalanced data] -.-> B[Precision, recall and F1]
    B -.-> C[Confusion matrix]
    C -.-> D[Accuracy]
    C -.-> E[ROC curve and AUC]
    D -.-> F[Classification problems]
    D -.-> G[Cross-validation]
    G -.-> H[Grid and random search]
    H -.-> I[Hyperparameter tuning]
    I -.-> J[Optuna]
    I -.-> K[Model-based learning]
    J -.-> I
    H -.-> L[ML pipelines]
    L -.-> M[Saving models with pickle]
    M -.-> N[Deployment]
    N -.-> O[Software integration]
    N -.-> P[MLOps and cost]
    N -.-> Q[Beta and A/B testing]
    N -.-> R[Retraining]
    R -.-> S[Model drift]
    S -.-> T[Online learning]
    R -.-> U[Batch offline learning]
    U -.-> V[OOB score]
    V -.-> W[Random forest]
    W -.-> X[Regression metrics]
    X -.-> Y[Regression problems]
    Y -.-> Z[Simple linear regression]
  
```

Figure 7: Concept map: Evaluating, tuning and production

4. The Learning path

Key point: Before each Video, read the Notes it builds on.

Each row lists a Video's Concepts and the Notes to read first. Videos marked *coming* or *deferred* do not have a Note yet.

Video	Concepts	Read first	Note
1	Data mining, Machine learning	nothing	written
2	Artificial intelligence, Deep learning, Features, Machine learning, Neural networks, Symbolic AI and expert systems	nothing	written
3	Anomaly detection, Association rule learning, Classification problems, Clustering, Dimensionality reduction, Labelled data, PCA, Regression problems, Reinforcement learning, Semi-supervised learning, Supervised learning, Unsupervised learning	Note 2	written
4	Batch (offline) learning, Deployment, Model drift, Retraining	nothing	written
5	Online learning, Out-of-core learning, Retraining, Stochastic gradient descent	Note 4	written

Video	Concepts	Read first	Note
6	Feature scaling, Instance-based learning, K-nearest neighbours,	Note 3	written
7	Model-based learning APIs, Deployment, Enough data, Feature construction and splitting, Feature engineering, Labelled data, MLOps and cost, Missing values, Outliers, Overfitting, Poor-quality data, Sampling noise and bias, Software integration, Underfitting, Web scraping	Note 2	written
8	Applications of ML	Note 2	deferred
9	Beta and A/B testing, Feature selection, Framing an ML problem, Hyperparameter tuning, Imbalanced data, ML development life cycle	Note 3 , Note 7	deferred
11	Features, One-hot encoding, Tensors	nothing	written
12	Anaconda, Jupyter and Colab	nothing	deferred

Video	Concepts	Read first	Note
13	Accuracy, CSV files, Data leakage, Deployment, Exploratory data analysis, Feature scaling, Feature selection, Logistic regression, ML development life cycle, ML pipelines, Saving models with pickle, Standardization, Train-test split	Note 3 , Note 6 , Note 7 , Video 9, coming	written
14	Framing an ML problem	nothing	deferred
15	CSV files	nothing	written
16	JSON and SQL data	nothing	written
17	APIs	Note 16	written
18	Web scraping	nothing	written
19	Correlation, Descriptive statistics, Exploratory data analysis, Variance	Note 15	written
20	Outliers, Skewness, Univariate analysis	Note 7 , Note 19	written
21	Bivariate and multivariate analysis, Correlation	Note 19 , Note 20	written
22	Pandas Profiling	nothing	written

Video	Concepts	Read first	Note
23	Binning and binarization, Encoding categorical data, Feature construction and splitting, Feature engineering, Feature extraction, Feature scaling, Feature selection, Feature transformation, Missing values, One-hot encoding, Outliers, Simple imputation (mean, median, mode, constant)	Note 11 , Note 13 , Note 20 , Note 21	written
24	Feature scaling, Standardization	Note 13 , Note 19 , Note 23	written
25	Normalization	Note 24	written
26	Encoding categorical data, Ordinal and label encoding	Note 13 , Note 23	written
27	Multicollinearity, One-hot encoding	Note 26	written
28	Column transformer	Note 23 , Note 26	written
29	Cross-validation, Deployment, Grid and random search, ML pipelines, Saving models with pickle	Note 7 , Video 9, coming, Note 13 , Note 28	written
30	Cross-validation, Function transformer, Q-Q plot	Note 20 , Note 23	written
31	Power transformer	Note 20 , Note 23 , Note 30	written
32	Binning and binarization	Note 23	written
33	Mixed variables	Note 23	written
34	Date and time features	Note 15 , Note 23	written

Video	Concepts	Read first	Note
35	Complete case analysis, Missing values	Note 7 , Note 20	written
36	Missing values, Simple imputation (mean, median, mode, constant)	Note 7 , Note 13	written
37	Missing values, Simple imputation (mean, median, mode, constant)	Note 7 , Note 13	written
38	Missing indicator	Note 37	coming
39	KNN imputer	Note 6 , Note 37	coming
40	Iterative imputation (MICE)	Note 37	coming
41	Outliers	Note 7 , Note 20	coming
42	Z-score outlier method	Note 19 , Video 41, coming	coming
43	IQR outlier method	Note 19 , Video 41, coming	coming
44	Percentile outlier method	Video 41, coming	coming
45	Feature construction and splitting	Note 23	coming
46	Curse of dimensionality, Dimensionality reduction, Feature extraction, Feature selection	Note 3 , Note 21 , Note 23	written
47	Feature extraction, PCA, Variance	Note 19 , Note 23 , Note 24 , Note 46	written
48	Covariance and covariance matrix, Eigenvectors and eigenvalues, PCA	Note 3 , Note 24 , Note 46 , Note 47	written
49	PCA	Note 24 , Note 46 , Note 47 , Note 48	written
50	Best-fit line and squared error, Simple linear regression	Note 3 , Note 6 , Note 27	written

Video	Concepts	Read first	Note
51	Best-fit line and squared error, Simple linear regression	Note 3 , Note 6 , Note 27	coming
52	Regression metrics	nothing	coming
53	Multiple linear regression	Video 51, coming	coming
54	Multiple linear regression	Video 51, coming	coming
55	Multiple linear regression	Video 51, coming	coming
56	Assumptions of linear regression	nothing	coming
57	Gradient descent	Note 24	coming
58	Batch gradient descent	Video 57, coming	coming
59	Stochastic gradient descent	Video 57, coming	coming
60	Mini-batch gradient descent	Video 57, coming	coming
61	Polynomial features, Polynomial regression	Video 55, coming	coming
62	Bias-variance trade-off	Note 7	coming
63	Regularisation, Ridge regression	Note 7 , Video 55, coming	coming
64	Ridge regression	Video 55, coming, Video 63, coming	coming
65	Ridge regression	Video 55, coming, Video 63, coming	coming
66	Ridge regression	Video 55, coming, Video 63, coming	coming
67	Lasso regression	Video 63, coming	coming
68	Lasso regression	Video 63, coming	coming
69	ElasticNet	Video 63, coming	coming
70	Logistic regression	Note 3 , Note 6 , Video 57, coming, Video 61, coming	coming
71	Logistic regression	Note 3 , Note 6 , Video 57, coming, Video 61, coming	coming

Video	Concepts	Read first	Note
72	Logistic regression	Note 3 , Note 6 , Video 57, coming, Video 61, coming	coming
73	Logistic regression	Note 3 , Note 6 , Video 57, coming, Video 61, coming	coming
74	Sigmoid function	nothing	coming
75	Logistic regression	Note 6 , Video 57, coming, Video 61, coming, Video 74, coming	coming
76	Accuracy, Confusion matrix	Note 13	coming
77	Precision, recall and F1	Video 9, coming, Video 76, coming	coming
78	ROC curve and AUC	Video 76, coming	coming
79	Softmax regression	Video 75, coming	coming
80	Polynomial features	nothing	coming
81	Hyperparameter tuning	nothing	coming
82	Naive Bayes	Note 3	coming
83	Naive Bayes	Note 3	coming
84	Naive Bayes	Note 3	coming
85	Naive Bayes	Note 3	coming
86	Naive Bayes	Note 3	coming
87	Naive Bayes	Note 3	coming
88	Naive Bayes	Note 3	coming
89	Naive Bayes	Note 3	coming
90	Naive Bayes	Note 3	coming
91	K-nearest neighbours	Note 3 , Note 6 , Note 24	coming
92	Support vector machines	Note 3	coming
93	Support vector machines	Note 3	coming
94	Support vector machines	Note 3	coming
95	Kernel trick	nothing	coming
96	Kernel trick	nothing	coming
97	Decision trees	Note 6	coming
98	Decision trees, Hyperparameter tuning	Note 6	coming

Video	Concepts	Read first	Note
99	Regression trees	Video 98, coming	coming
100	Decision trees	Note 6	coming
101	Ensemble learning	Video 100, coming	coming
102	Voting ensembles	Video 101, coming	coming
103	Voting ensembles	Video 101, coming	coming
104	Voting ensembles	Video 101, coming	coming
105	Bagging	Note 7 , Video 101, coming	coming
106	Bagging	Note 7 , Video 101, coming	coming
107	Bagging	Note 7 , Video 101, coming	coming
108	Random forest	Video 100, coming, Video 107, coming	coming
109	Bias-variance trade-off, Random forest	Note 7 , Video 100, coming, Video 107, coming	coming
110	Random forest	Video 100, coming, Video 107, coming	coming
111	Hyperparameter tuning, Random forest	Video 100, coming, Video 107, coming	coming
112	Cross-validation, Grid and random search	Note 29 , Video 111, coming	coming
113	OOB score	nothing	coming
114	Feature importance	Video 111, coming	coming
115	AdaBoost	nothing	coming
116	AdaBoost	nothing	coming
117	AdaBoost	nothing	coming
118	AdaBoost, Hyperparameter tuning	nothing	coming
119	Boosting	Video 101, coming	coming
120	Gradient boosting	Video 57, coming, Video 119, coming	coming
121	Gradient boosting	Video 57, coming, Video 119, coming	coming
122	Gradient boosting	Video 57, coming, Video 119, coming	coming
123	XGBoost	Video 122, coming	coming
124	XGBoost	Video 122, coming	coming
125	XGBoost	Video 122, coming	coming

Video	Concepts	Read first	Note
126	XGBoost	Video 122, coming	coming
127	Stacking and blending	Video 101, coming	coming
128	Clustering, K-means	Note 3 , Note 24	coming
129	K-means	Note 24 , Video 128, coming	coming
130	K-means	Note 24 , Video 128, coming	coming
131	Hierarchical clustering	Video 128, coming	coming
132	DBSCAN	Video 128, coming	coming
133	Imbalanced data	nothing	coming
134	Optuna	Video 118, coming	coming

5. The Algorithm chooser

Key point: Two questions narrow the choice: do we have labelled outputs, and are they numbers or categories?

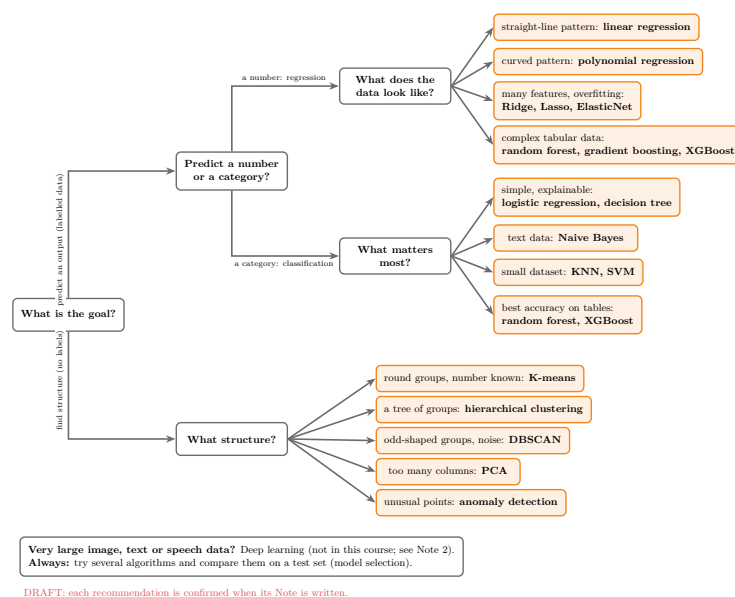


Figure 8: The Algorithm chooser (draft)

Figure 8 is a starting point, not a rule: in practice we try several suitable algorithms and compare them on a test set. It is a draft until the algorithm Notes are written.

6. Key terms

Term	Meaning
Course map	The overview that ties all Notes together, in four views
Pipeline map	Where each Concept sits among the 14 steps of an ML project
Concept map	How Concepts are connected by Links
Learning path	Which Notes to read before which
Algorithm chooser	A flowchart from a problem's properties to suitable algorithms
Concept	One idea on the Course map
Link	A labelled connection between two Concepts
Draft / confirmed	Guessed from titles / checked against a written Note